

Project Design Phase
Solution Architecture

Date	20-02-2026
Team ID	LTVIP2026TMIDS24662
Project Name	<i>Shopsmart: Your Digital Grocery</i>
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for **ShopSmart** ensures a **scalable, secure, and user-friendly platform** that connects customers with **local grocery stores**. The system is built using the **MERN stack** (MongoDB, Express.js, React.js, Node.js) and includes **separate modules** for **user, vendor (store), and admin** roles.

The architecture emphasizes:

- Seamless grocery shopping and store discovery
- Real-time cart, inventory, and order management
- Secure login for users, vendors, and admins with role-based routing
- Scalable and modular backend APIs built using RESTful principles
- Structured data flow between the React frontend, Node backend, and MongoDB
- Conditional rendering and access control based on user roles (user/vendor/admin)
- Admin tools for promoting stores, managing users, and maintaining platform quality

Example - Solution Architecture Diagram:

Solution Architecture

ShopSmart

